

Anirudh Tyagi

✉ anirudhtyagi188@gmail.com ☎ +91 73032 89966 in linkedin.com/in/itsanirudhtyagi 🌐 anirudh-tyagi

Education

Sai University

Sept 2022 – May 2026

B.Tech in Computing and Data Science (CGPA - 8.0)

- **Relevant Coursework:** Machine Learning, Deep Learning, Reinforcement Learning, Artificial Intelligence, Data Structures & Algorithms, Speech and Image Analysis, Bio Informatics, Software Engineering, DBMS, Data Analysis and Visualization

Experience

Software Engineer Intern

Noida, India

Cadence Design Systems

Jan 2026 – Present

- Developed production-grade features for Allegro System Capture, a large-scale C++/Qt EDA application in the Allegro R&D team, leveraging modern C++, STL, and OOP principles, delivered enhancements including a real-time schematic world-view widget with live viewport synchronization and an overhauled docking framework with deferred state restoration, orientation-aware resizing, and context-aware toolbar visibility for stable multi-workspace behavior
- Engineered a tamper-proof expiry validation module for early-access builds using asymmetric digital signatures and authenticated encryption to enforce version-bound expiry controls, ensuring only cryptographically verified license files are accepted at runtime; delivered <10ms end-to-end validation latency alongside a companion CLI for generating and auditing signed license artifacts

Software Engineer Intern

Noida, India

Cadence Design Systems

June 2025 – Aug 2025

- Worked as part of the Allegro R&D team, developed data acquisition pipelines and backend infrastructure using Python and Flask for scalable ML workflows
- Optimized ML model achieving 91% accuracy, ensured efficient and secure inference delivery

Research Intern

Chennai, India

National Institute of Ocean Technology

July 2024 – Sept 2024

- Processed and analyzed 18 TB of deep-sea metagenomic data through a full bioinformatics pipeline of de novo assembly, BLAST homology search, and multi-sequence alignment, uncovering novel microbial genes and species with potential industrial biotechnology applications

Skills

Languages: C++, C, Python, JavaScript, TypeScript, HTML, SQL, R, Julia, Tel

Frameworks & Libraries: Django, Flask, Node.js, React.js, PyTorch, TensorFlow, Scikit-learn, OpenCV

Tools & Platforms: Git, Perforce, AWS, Docker, Hugging Face, Jupyter Notebook, PyCharm, VS Professional, Jira, RHEL

AI/ML & Domains: Machine Learning, Deep Learning, NLP, Computer Vision, Reinforcement Learning, Speech & Image Processing, Retrieval-Augmented Generation (RAG), Predictive Modeling, Statistical Analysis, Data Analytics, Data Preprocessing

Projects

Smart Home Energy AI(*Python, PyTorch, Scikit-learn, Reinforcement Learning, Flask, Gymnasium, SQLAlchemy*)

[link-git](#) 

- Engineered a DQN-based energy management system integrating 3 supervised ML models within a custom Gymnasium RL environment to optimize HVAC and lighting control, achieving 26.4% simulated energy reduction, 98% comfort rating, and 97.8% occupancy prediction accuracy at sub-15ms inference latency; built a Flask/SocketIO dashboard synchronizing 50+ IoT streams into a closed-loop AI decision engine with 200ms end-to-end response time

Cyber Cipher(*RESTful API Development, Python, React.js, TypeScript, Tailwind*)

[vercel-app](#) 

- Built a full-stack cryptographic analysis platform integrating RC4 and ChaCha20 cipher engines with statistical security analysis modules for encryption strength evaluation, featuring a responsive React/TypeScript UI with real-time metric visualizations served via a Python RESTful API

Credit Approval System(*Django, Django REST Framework, PostgreSQL, Celery, Docker*)

[link-git](#) 

- Engineered a production-ready loan management backend with RESTful APIs covering the full credit lifecycle with customer registration, credit scoring, eligibility assessment, and loan approval with Celery async processing, Docker containerization, and 100% pytest coverage across all endpoints

Research & Publications

- Behl D., Tyagi A., Kandhoul N., Dhurandher S. “Post Quantum Federated learning and Blockchain with Resilient IPFS for Cyber logs (FABRIC)”, under review at *Journal of Network and Systems Management*, Springer.

Leadership & Activities

Cadence India Hackathon

2025

- Presented BLINC, a decentralized log analysis system integrating federated learning, blockchain (PoS), and IPFS

Sports Society President, Sai University

2022-2023